

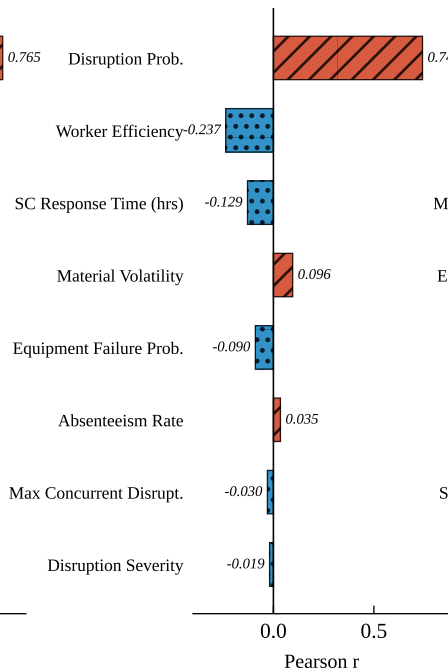
(a)

S1: Baseline



(b)

S2: Centralized BIM+DT



(c)

S3: Tokenized BIM-Twin



Positive r (risk factor) Negative r (protective)